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BFM Module B Concept Unit wise marathon :

Risk Management

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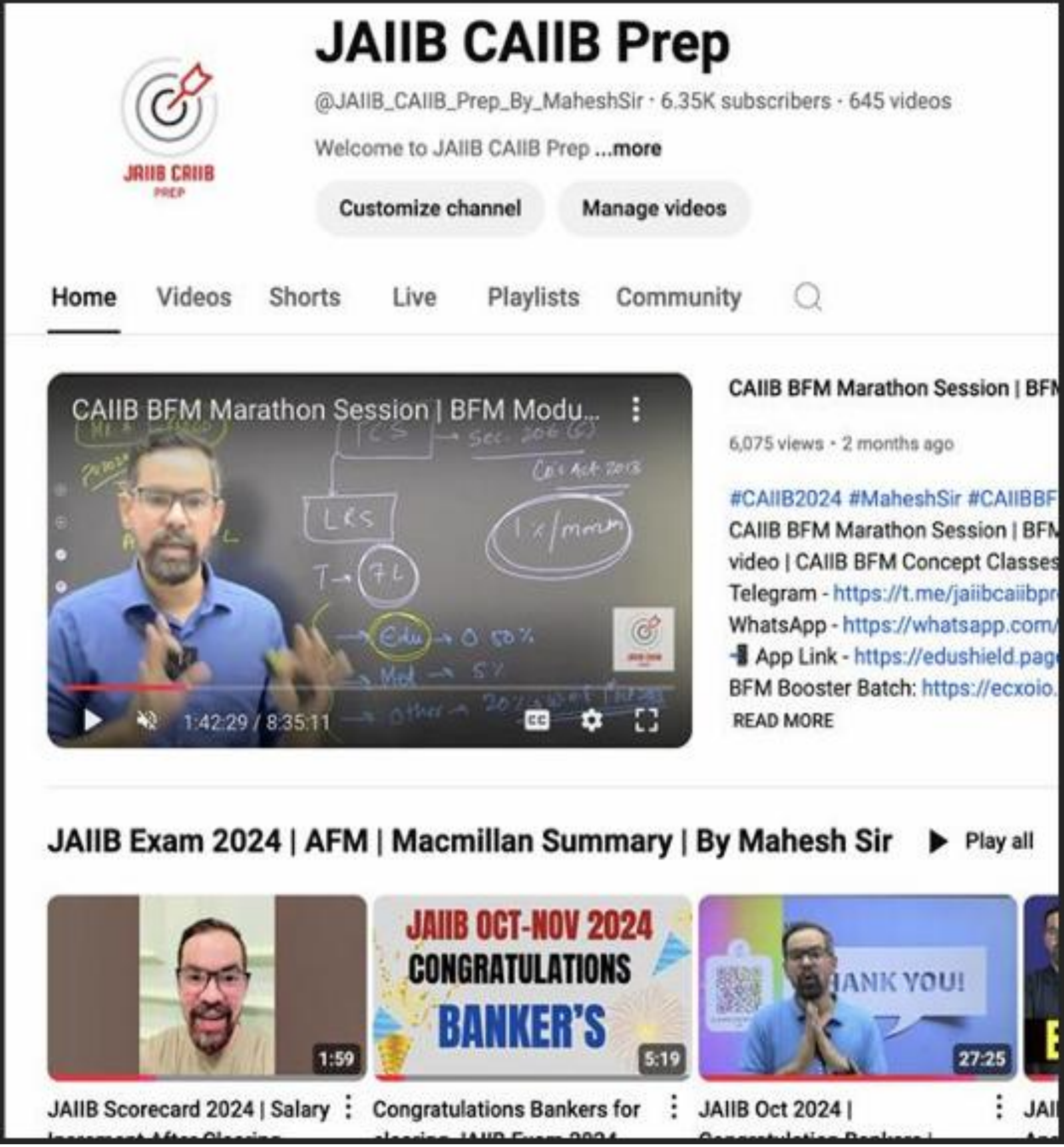

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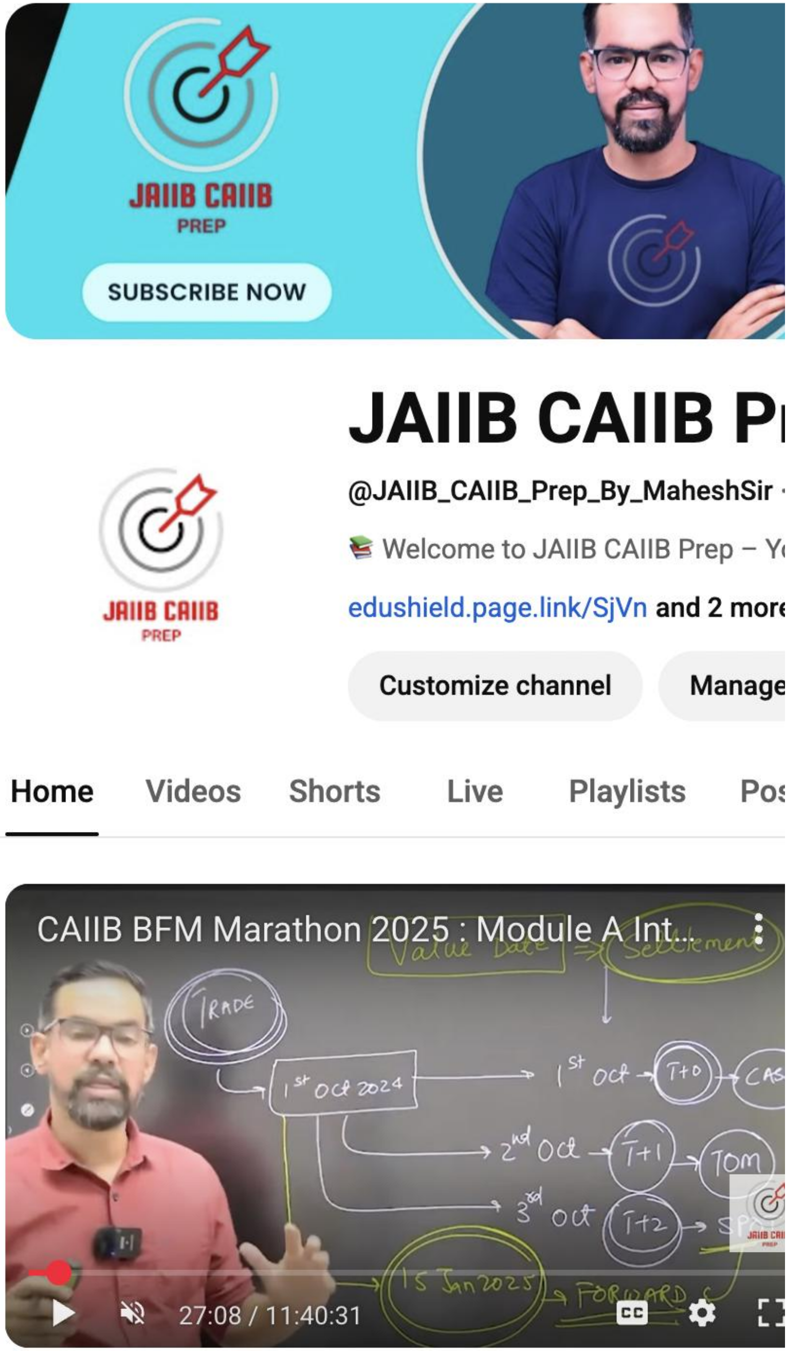

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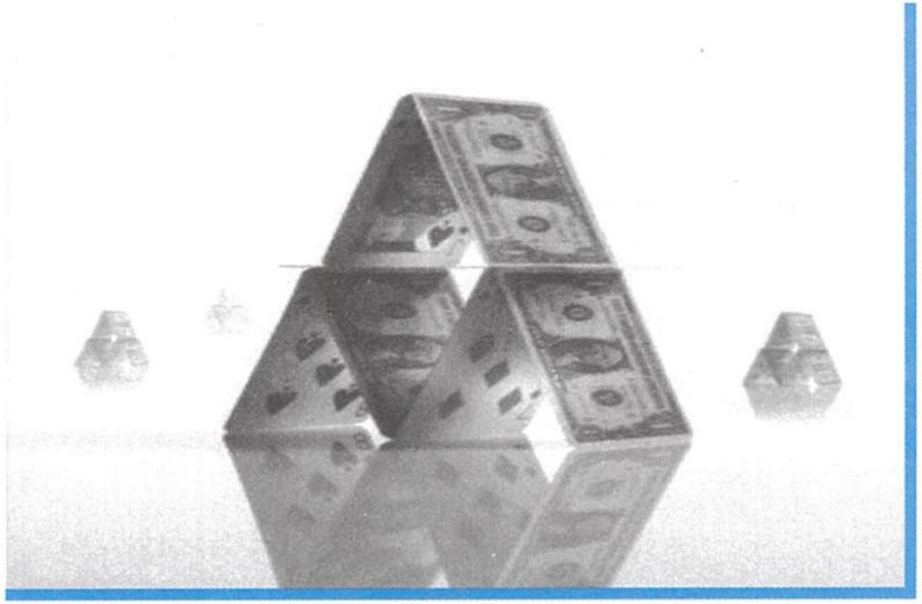
MODULE B



RISK MANAGEMENT

UNIT 11

Risk and Basic Risk Management Framework

STRUCTURE

Risk, Capital & Return

- Higher the risk, higher the capital & vice- versa.
- Bank maintain capital funds as percentage of RWAs.
- Banks creating low risk assets can maintain low capital
- Bank exposing to high risk assets, have to maintain high capital

Risk Management

Response to these considerations calls for risk management framework in an organization that has well articulated processes covering the following areas:

- Organization for Risk Management
- Risk Identification
- Risk Measurement
- Risk Pricing
- Risk Monitoring and Control
- Risk Mitigation

11.4.1 Organisation for Risk Management

Usually, risk management organization consists of:

- The Board of Directors
- The Risk Management Committee of the Board
- The Committee of senior-level executives, also called as Sub-Groups:
    1. Asset-Liability Committee (ALCO)
    2. Credit Risk Management Committee (CRMC)
    3. Operational Risk Management Committee (ORMC)
- The Risk management support group/department

The Board of Directors has the overall responsibility for management of risks. The Board articulates risk management policies, procedures, aggregate risk limits, review mechanisms and reporting and auditing systems. The Board should decide the risk management policy of the bank and set limits for various risks.

The Risk Management Committee is a Board level Sub-Committee. The responsibilities of Risk Management Committee with regard to risk management aspects include the following:

- Setting guidelines for risk management and reporting
- Ensure that risk management processes conform to the policy
- Setting up prudential limits and their periodical review
- Ensure robustness of risk measurement models
- Ensure proper manning for the processes

1. ALCO:
    a. Oversees the Liquidity risk of the Bank.
    b. Monitors and tries to protect/improve the NIM of the Bank.
    c. Recommends BPLR / Base Rate / MCLR/RLLR of the Bank to the Board.
    d. Mid Office Group:
    e. Monitoring of Treasury Operations on real time basis.
2. CRMC:
    a. Ensuring robustness of the Credit Risk in the Bank including Credit Rating processes/models etc.
    b. Working out:
        i. Yearly Rating-wise Distribution Charts.
        ii. Yearly Rating-wise Migration Charts.
        iii. Yearly Probability of Default (PD) for individual ratings as well as average PD.
    c. Conducting Portfolio analysis for retail loan assets.

3. ORMC:
   a. Ensuring the robustness of Operational risk in the Bank.
   b. Analysis of the Frauds in the Bank and comes out with risk mitigating measures for non-recurrence of such nature.
   c. Clears any new products / processes (other than credit) put up by business units.

Risk Management Support Group/Department:

(i) The Risk Management Department will be headed by a Chief Risk Office Officer – CRO (usually in the cadre of General Manager in Public Sector Banks).

(ii) He will have complement of qualified / skilled staff to manage the following Departments:

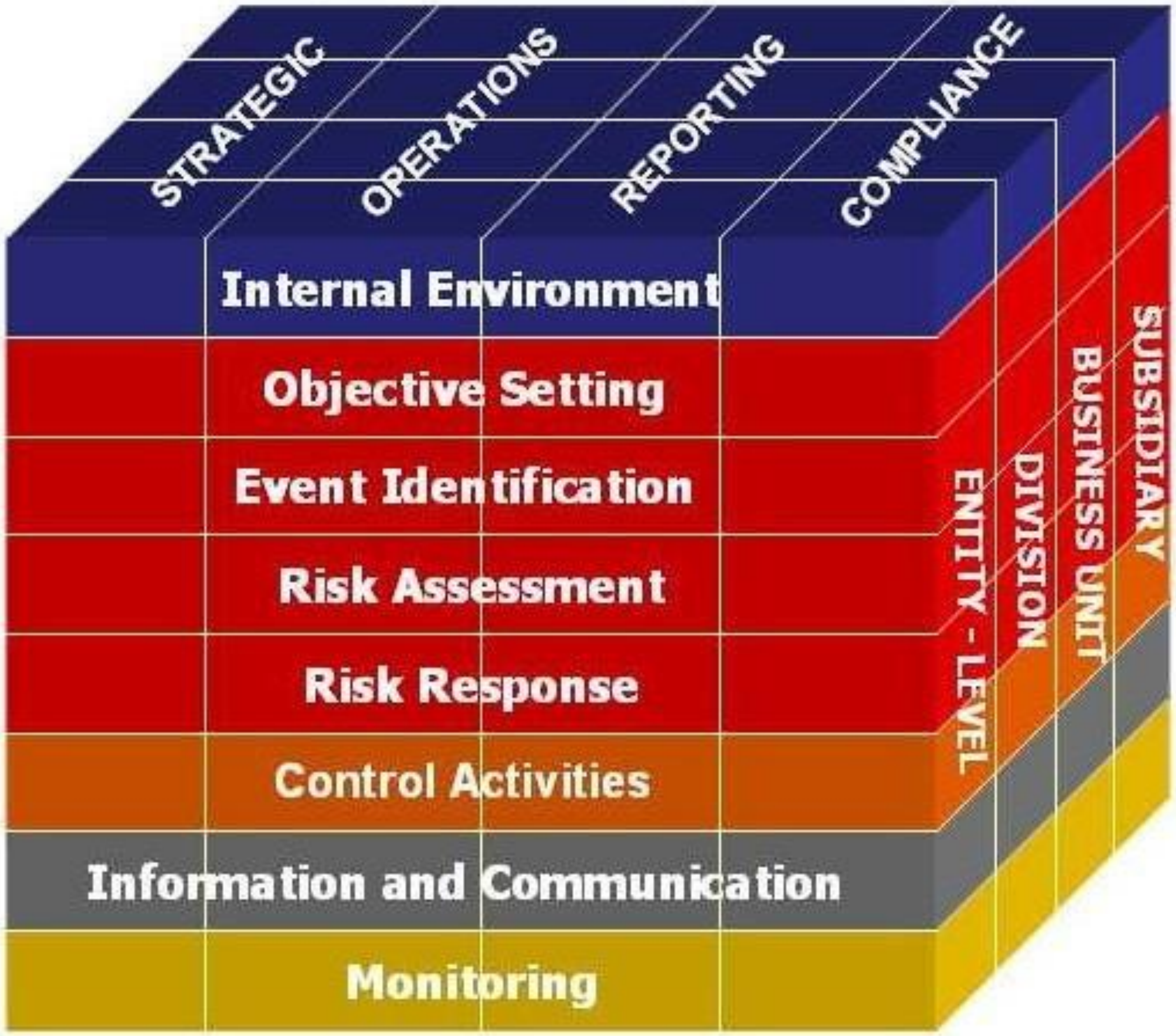
- Asset/Liability Group to look into the Market Risk (Liquidity, Interest rate and Forex)
- Mid Office Group to monitor the Treasury Operations.
- Credit risk Group to monitor the Credit Risk.
- Operational risk Group to monitor the Operational and Residual risk.

Risk Management

Response to these considerations calls for risk management framework in an organization that has well articulated processes covering the following areas:

- Organization for Risk Management
- Risk Identification
- Risk Measurement
- Risk Pricing
- Risk Monitoring and Control
- Risk Mitigation

The ERM Framework



EWRM involves the following steps:

- Review of the internal environment with a view to assess the risk philosophy and risk culture
- Review of the process of setting objectives
- Assessment of the existing mechanism of identifying risk-factors that can affect achievement of the desired objectives
- Evaluation of the existing process of assessing risks
- Assessment of the process of responding to identified risks
- Evaluation of the adequacy of existing control processes
- Assessment of the adequacy of existing management information system (MIS)
- Review of the process of monitoring risks

EWRM involves the following steps:

Banks have identified and started adapting the Enterprise Risk Management Framework released **by COSO (Committee of Sponsoring Organizations of the Treadway Commission)** as a framework to drive their initiatives in risk management beyond Basel norms and regulatory compliances.

The COSO ERM framework has all the components that could help the banks to stand a chance to derive business value while meeting compliance requirements.

Eight key components viz Internal Environment, Objective setting, Risk Assessment, Risk response, Control Activities, Information & Communication, Monitoring

Four key objectives of business viz. strategic, operations, reporting and compliance.

UNIT 12

Risks in Banking Business

STRUCTURE

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Risk Identification In Banking Business

Banking business lines are many and varied. Commercial banking, corporate finance, retail banking, trading and investment banking and various financial services form the main business lines of banks Within each line of business, there are sub-groups and each sub-group contains a variety of financial activities.

Bank's clients may vary from retail consumer segment to mid-market corporate/large corporate financial institutions.

Banking services may differ appreciably for each segment even for similar services.

Table 12.1 Business Lines of the Banking Industry

Business Lines	*Sub-groups*	*Activities*
Corporate Finance	Corporate Finance	Mergers and acquisitions, underwriting, privatisations, securitisation, research, debt (government, high yield), equity, syndications, IPO, secondary private placements
	Municipal/Government Finance	
	Merchant Banking	
	Advisory Services	
Trading and Sales	Sales	Fixed income, equity, foreign exchanges, commodities, credit, funding, own position securities, lending and repos, brokerage, debt, prime brokerage
	Market making	
	Proprietary Positions	
	Treasury	
Retail Banking	Retail Banking	Retail Lending & Deposits, Banking Services, Trust and Estates
	Private Banking	Private Lending & Deposits, Banking Services,Trust and Estates, Investment Advice.
	Card Services	Merchant/Commercial/Corporate Cards, PrivateLabels and Retail
Commercial Banking	Commercial Banking	Project finance, real estate, export finance, trade finance, factoring, leasing, lending, guarantees,bills of ex

Business Lines	*Sub-groups*	*Activities*
Payments and Settlement	External Clients	Payments and collections, funds transfer, clearing and settlement
Agency Services	Custody	Escrow, depository receipts, securities lending (customers), corporate actions
	Corporate Agency	Issuer and paying agents
	Corporate Trust	
Asset Management	Discretionary Fund Management	Pooled, segregated, retail, institutional, closed,
	Non-Discretionary Fund	open, private equity
	Management	Pooled, segregated, retail, institutional, closed, open
Retail Brokerage	Retail brokerage	Execution and full service

From the risk management point of view, banking business lines may be grouped broadly under the following major heads.

- The Banking Book
- The Trading Portfolio
- Off-Balance Sheet Exposures

The Banking Book

The banking book includes all advances, deposits and borrowings, which usually arise from commercial and retail banking operations.

All assets and liabilities in banking book have the following characteristics:

- **They are normally held until maturity.**
- **Accrual system of accounting is applied.**
- **They are not subjected to MTM (mark to market) exercise.**
- **They attract capital charge on credit risk and not on market risk.**

The Trading Book

The trading book mostly comprises of fixed income securities, equities, foreign exchange holdings, commodities, etc., held by the bank on its own account.
The trading book includes all the assets that are marketable, i.e., they can be traded in the market.

Contrary to the characteristics of assets and liabilities held in the banking book, the trading book assets have the following characteristics:

- They are normally not held until maturity and positions are liquidated in the market after holding the assets for a certain period

- Mark-to-Market system is followed and the difference between the market price and the book value is taken to profit and loss account.

Off-Balance Sheet Exposures

Off-balance sheet exposures are contingent in nature. Where banks issue guarantees, committed or backup credit lines, letters of credit, etc., banks face payment obligations contingent upon some event.

- ✓ These contingencies adversely affect the revenue generation of banks. Banks may also have contingent assets (for example, a bank may have purchased insurance to protect against certain negative events). Here banks are the beneficiaries subject to certain contingencies.
- ✓ Derivatives are off-balance sheet market exposures. They may be swaps, futures, forward contracts, foreign currency contracts, options, etc.

Banking Risks – Definitions

From the discussion above, we may summarise the major risks in banking business or "Banking Risks'. They are listed as follows:

- **Liquidity Risk**
- **Interest Rate Risk**
- **Market Risk**
- **Default or Credit Risk**
- **Operational Risk**

12.5.6 Model Risk

Models are designed to predict values of variables for which it is specifically designed. Value of a given variable would depend upon one or more parameters, which influence the value of the given variable.

Model risk usually arises because of the following reasons:

- Incorrect assumptions or assumptions, which have become non-relevant
- Ignoring one or more parameters – usually for simplification or for some practical reasons
- Errors of statistical techniques or insufficient data points
- Incorrect judgment in dealing with outliers, etc.

Risks faced by banking and financial services may be summarised as shown in Fig. 12.1.

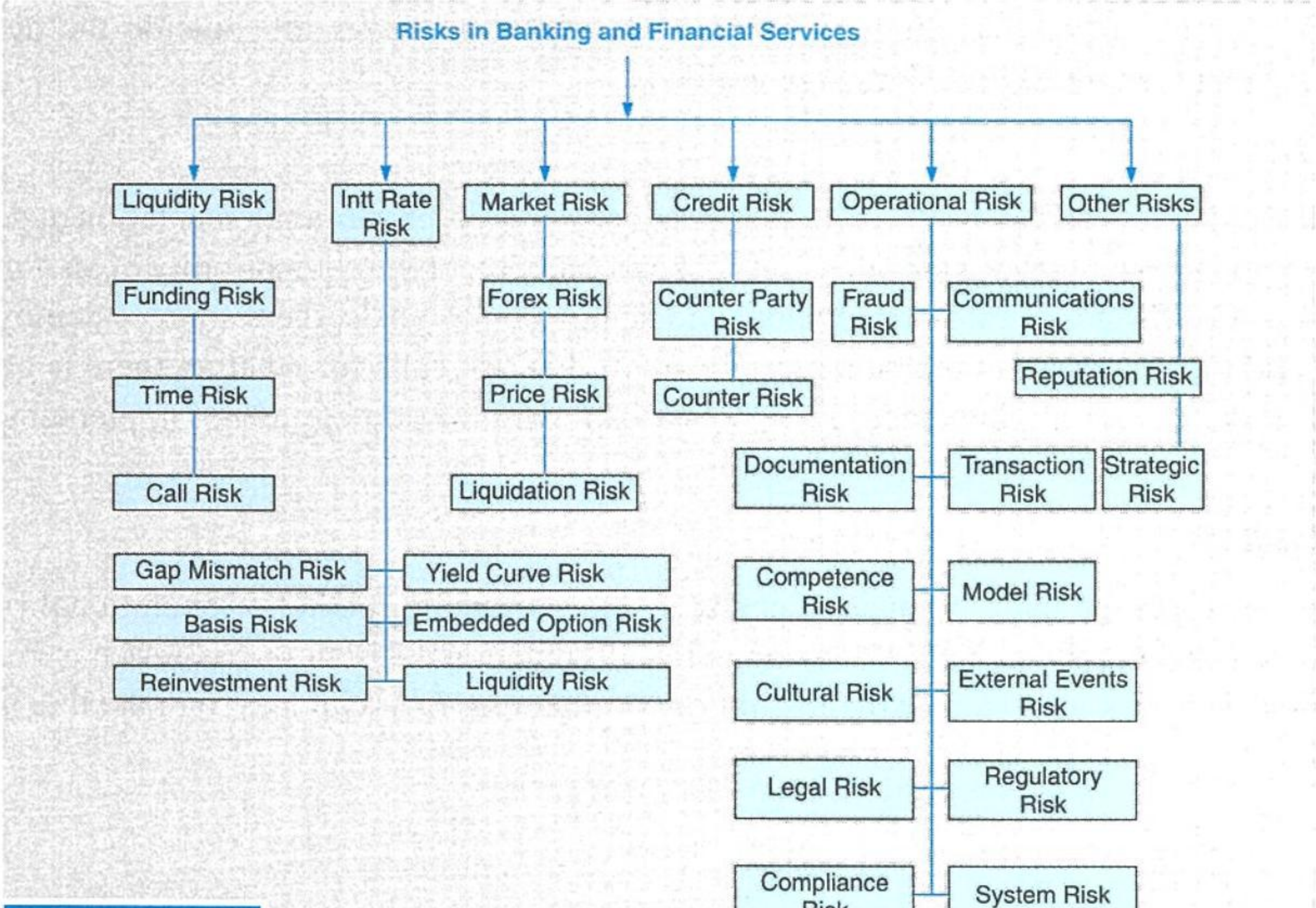

FIGURE 12.1

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UNIT 13 Risk Regulations in Banking industry

STRUCTURE

Regulation of Banking Industries - Necessities And Goals

Banking and financial services, all over the world, are regulated usually by the Monetary Authority.

This is because banking and financial services are the backbone of an economy.

A healthy and strong banking system is a must for any economy to function smoothly and to prosper. As we have seen, banks have risks and risk taking is their business.

But if risk-taking is not regulated properly, banks may fail and it would have a disastrous effect on the economy. Therefore, Monetary Authorities across the world **regulate functioning of the banks**. In India, this function, as we all know, is with Reserve Bank of India, Country's monetary authority.

Regulations have several goals. They are:

- Improving the safety of the banking industry, by imposing capital requirements in line with bank's risks.
- Levelling the competitive playing field of banks through setting common benchmarks for all players.
- Promoting sound business and supervisory practices. Controlling and monitoring Systemic Risk’.
- Protecting interest of depositors as depositors cannot impose a real market discipline on banks.

Systemic Risk

Systemic risk is the risk of failure of the whole banking system. An Individual bank's failure is one of the major sources of the systemic risk.

This happens because of high inter-relations that exist on a ongoing basis between banks through mutual lending and borrowing and other commitments.

Basel Committee on Banking Supervision

On 26th June 1974, a number of banks had released Deutschmarks to Bank Herstatt in Frankfurt in exchange for dollar payments that were to be delivered in New York.

Due to differences in time zones, there was a lag in dollar payments to counterparty banks during which Bank Herstatt was liquidated by German regulators (Bundesbank), i.e., before the dollar payments could be effected.

- **Policy Development Group**
- **Supervision and Implementation Group**
- **Basel Consultative Group**
- **Macro prudential Supervision Group**
- **Accounting Experts Group**

The Basel I

- In 1988, The Basel Committee on Banking Supervision (BCBS) introduced capital measurement system called Basel capital accord, also called as Basel 1.
- If focused almost entirely on credit risk, it defined capital and structure of risk weight for banks.
- The minimum capital requirement was fixed at **8% of risk-weighted assets (RWA).**
- India adopted **Basel 1 guidelines in 1999.**
- In India, however banks are required to maintain a minimum Capital-to-risk weighted Asset ratio (CRAR) of **9% on an ongoing basis.**

1996 Amendment in Basel I

The 1996 Market Risk Amendment introduced an explicit capital cushion for the market risks to which banks are exposed because of their trading in equities, debt securities, foreign exchange, commodities, options, etc. Allowed banks to use their internal rating models to estimate their market risks.

13.4 1996 AMENDMENT TO INCLUDE MARKET RISK

In 1996, BCBS published an amendment to the 1988 Basel Accord to provide an explicit capital cushion for the price risks to which banks are exposed, particularly those arising from their trading activities. This amendment was brought into effect in 1998.

Salient features of the amendment are given below:

1. Allows banks to use proprietary in-house models for measuring market risks.
2. Banks using proprietary models must compute VAR daily, using a 99th percentile, one-tailed confidence interval with a time horizon of ten trading days using a historical observation period of at least one year.
3. The capital charge for a bank that uses a proprietary model will be the higher of the previous day's VAR and three times the average of the daily VAR of the preceding 60 business days.
4. Use of 'back-testing' (ex-post comparisons between model results and actual performance, which is also called a model validation) to arrive at the 'plus factor' that is added to the multiplication factor of three.
5. Allows banks to issue short-term subordinated debt subject to a lock-in clause (Tier 3 capital) to meet a part of their market risks.
6. Alternate standardized approach using the 'building block' approach where general market risk and specific security risk are calculated separately and added up.
7. Banks to segregate trading book and mark to market all portfolio/position in the trading book.
8. Applicable to both trading activities of banks and non-banking securities firms.

The Basel II

In 2004, Basel II guidelines were published by BCBS, which were considered to be the refined and reformed versions of **Basel I accord.**

Three Pillars of Basel II

- **Minimum capital Requirement**
- **Supervisory review process**
- **Market Discipline**

Minimum capital Requirement

The Capital base of the bank consist of the following three types of capital element. **Tier 1, Tier 2 and Tier 3 capital**. The sum of Tier 1, Tier 2 and Tier 3 element will be eligible for inclusion in the capital base, subject to the following limits.

(a)Total of Tier 2 (Supplementary) elements will be limited to a maximum of 100% of the Tier 1 element.

(b)Subordinated term debt will be limited to a maximum of 50% of Tier 1 elements.

(c)Tier 3 capital will be limited to 250% of a bank's Tier 1 capital that is required to support market risks.

(d)Where general provisions/general loan –loss reserves include amounts reflecting lower valuations of assets

(e)Asset revaluation reserves, which take the form of latent gains on unrealized securities, will be subject to a discount of 55%.

Supervisory review process

The section discusses the key principles of supervisory review, risk management guidance and supervisory transparency and accountability, produced by the committee with respect to banking risks.

Principles of Supervisory Review

Principle 1: Banks should have a process for assessing their overall capital adequacy in relation to their risk profile and a strategy for maintaining their capital levels.

Principle 2: Supervisors should review and evaluate Bank's internal capital adequacy assessments and strategies, as well as their ability to monitor and ensure their compliance with regulatory capital ratios. Supervisors should take appropriate supervisory action if they are not satisfied with the result of this process.

Principle 3: Supervisors should expect banks to operate above the minimum regulatory capital ratios and should have the ability to require banks to hold capital in excess of the minimum.

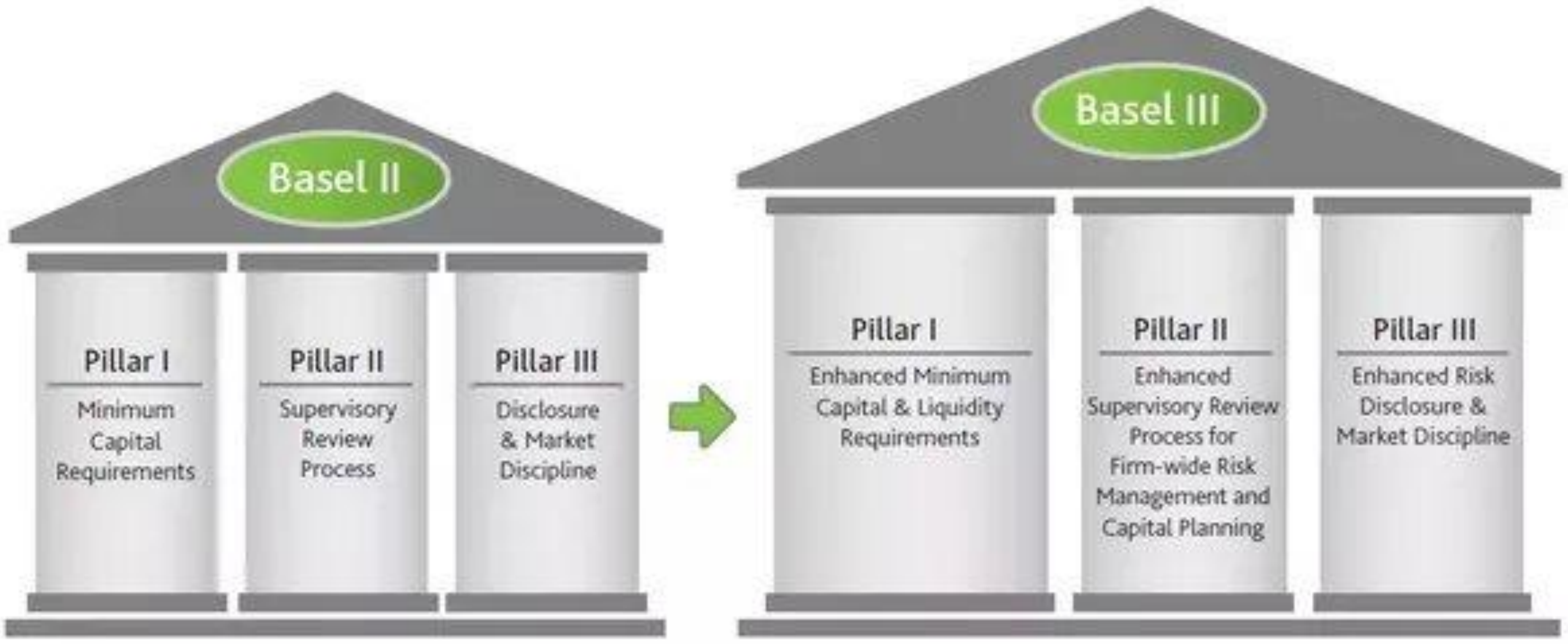
Principle 4: Supervisors should seek to intervene at an early stage to prevent capital from falling below the minimum levels required to support the risk characteristics of a particular bank and should require rapid remedial action if capital is not maintained or restored.

Market Discipline

- Disclosure Requirements
- Guiding Principles
- Achieving Appropriate Disclosure

Basel III

Basel III or Basel 3 released in December, 2010 is the third in the series of Basel Accords. These accords deal with risk management aspects for the banking sector. So we can say that **Basel III is the global regulatory standard on bank capital adequacy, stress testing and market liquidity risk.**


Basel II
Pillar I
Minimum Capital Requirements
Pillar II
Supervisory Review Process
Pillar III
Disclosure & Market Discipline
Basel III
Pillar I
Enhanced Minimum Capital & Liquidity Requirements
Pillar II
Enhanced Supervisory Review Process for Firm-wide Risk Management and Capital Planning
Pillar III
Enhanced Risk Disclosure & Market Discipline

Basel III

The RBI issued Guidelines based on the Basel III reforms on capital regulation on May 2 2012, to the extent applicable to banks operating in India.

The Basel III capital regulation has **been implemented form April 1, 2013** in India in phase and it will be fully implemented as on March 31, 2019 but Extended.

- **Improve the banking sector's ability to absorb ups and downs arising from financial and economic instability**
- **Improve risk management ability and governance of banking sector**
- **Strengthen banks' transparency and disclosures**

Better Capital Quality:

One of the key elements of Basel 3 is the introduction of much stricter definition of capital. Better quality capital means **the higher loss- absorbing capacity.**
This in turn will mean that banks will be stronger, allowing them to better withstand periods of stress.

Capital Conservation Buffer:

Another key feature of Basel III is that now banks will be required **to hold a capital conservation buffer of 2.5%**.
The aim of asking to build conservation buffer is to ensure that banks maintain a cushion of capital that can be used to absorb losses during periods of financial and economic stress.

Minimum Common Equity and Tier 1 Capital Requirements:

The minimum requirement for common equity, the highest form of loss-absorbing capital, has been raised under **Basel III from 2% to 4.5% of total risk-weighted** assets. The overall Tier 1 capital requirement, consisting of not only common equity but also other qualifying financial instruments, will also increase from the current minimum of 4% to 6%.

Leverage Ratio:

A review of the financial crisis of 2008 has indicted that the value of many assets fell quicker than assumed from historical experience. Thus, now Basel III rules include a leverage ratio to serve as a safety net.
This aims to put a cap on swelling of leverage in the banking sector on a global basis. 3% leverage ratio of Tier 1 will be tested before a mandatory leverage ratio is introduced in January 2018.

Liquidity Ratios:

Under Basel III, a framework for liquidity risk management will be created. A new Liquidity Coverage Ratio (LCR) and Net Stable Funding Ratio (NSFR) are to be introduced in 2015 and 2018, respectively.

Liquidity Coverage Ratio (LCR) Formula

$$\text{LCR Ratio} = \left(\frac{\text{Value of High-Quality Liquid Assets (HQLA)}}{\text{Total Net Cash Outflows Over the Next 30 Calendar Days}} \right) \text{X 100}$$

$$\left(\frac{Available\ stable\ funding}{Required\ stable\ funding} \right) \geq 100\%$$

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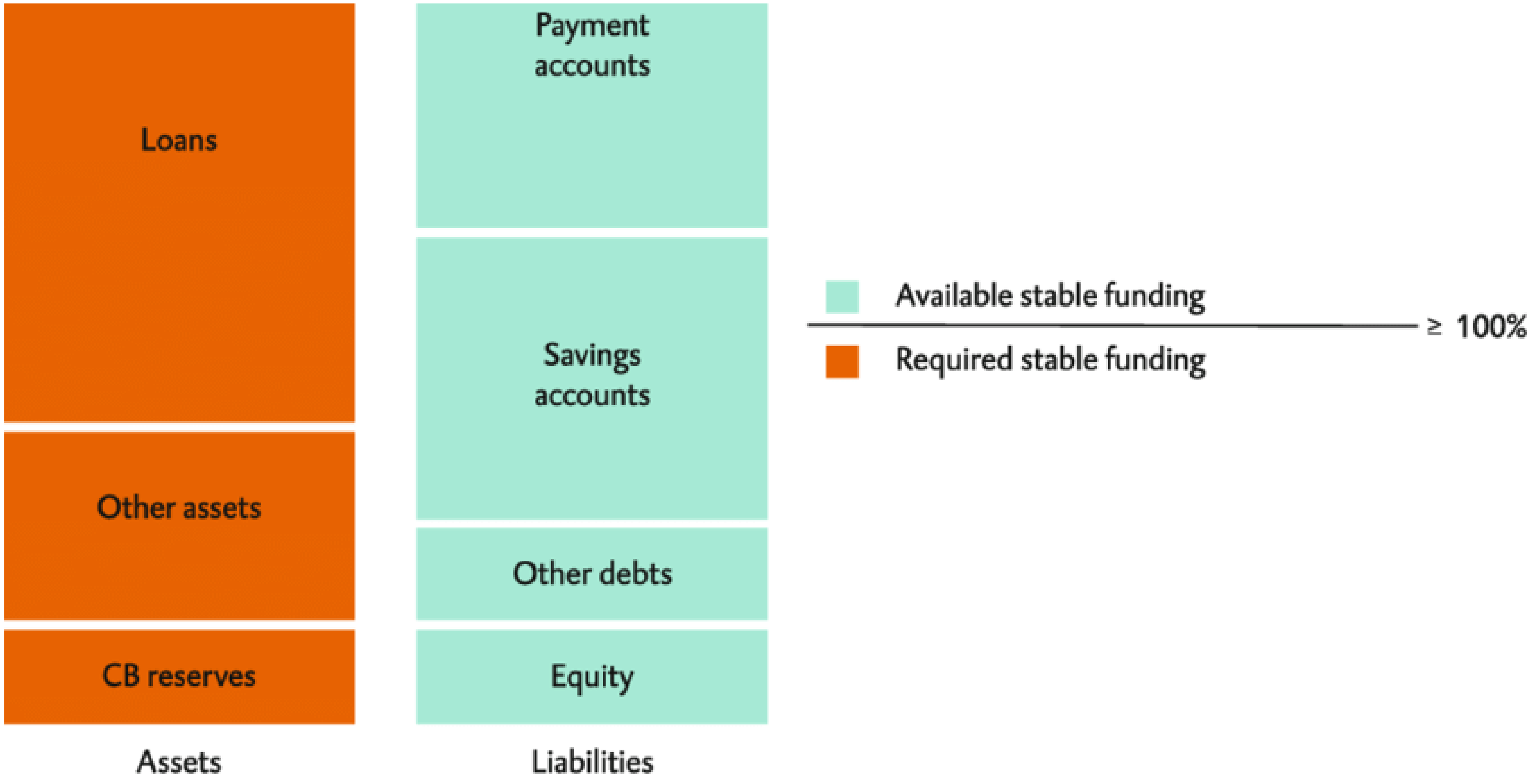

Liquidity Coverage Ratio (LCR) vs. Net Stable Funding Ratio (NSFR)

FACTORS	LCR	NSFR
Purpose	Liquidity Requirement	Funding Requirement
Time Frame	30 Days	12 Months
Assets Used	Liquid Assets	Both Liquid & Non-Liquid Assets

Systemically Important Financial Institutions (SIFI):

As part of the macro- prudential framework, systemically important banks will be expected to have loss-absorbing capability beyond the Basel III requirements.

Options for implementation include capital surcharges, contingent capital and bail-in-debt.

The list of D-SIBs is as follows:

Bucket	Banks	Additional Common Equity Tier 1 requirement as a percentage of Risk Weighted Assets (RWAs)
5	-	1%
4	State Bank of India*	0.80%
3	-	0.60%
2	HDFC Bank*	0.40%
1	ICICI Bank	0.20%

* The higher D-SIB surcharge for SBI and HDFC Bank will be applicable from April 1, 2025. Hence, up to March 31, 2025, the D-SIB surcharge applicable to SBI and HDFC Bank will be 0.60% and 0.20% respectively.

Capital Requirements under Basel III

Thus, with full implementation of capital ratios and CCB the capital requirements as on 1st October, 2021will be as follows:

S. No.	Regulatory Capital	As % to RWAs
(i)	Minimum Common Equity Tier 1 Ratio	5.5
(ii)	Capital Conservation Buffer (comprised of Common Equity)	2.5
(iii)	Minimum Common Equity Tier 1 Ratio plus Capital Conservation Buffer [(i)+(ii)]	8.0
(iv)	Additional Tier 1 Capital	1.5
(v)	Minimum Tier 1 Capital Ratio [(i) +(iv)]	7.0
(vi)	Tier 2 Capital	2.0
(vii)	Minimum Total Capital Ratio (MTC) [(v)+(vi)]	9.0
(viii)	Minimum Total Capital Ratio plus Capital Conservation Buffer [(vii)+(ii)]	11.5

Capital Charge for Credit Risk

Under the Standardised Approach, the rating assigned by the eligible external credit rating agencies will largely support the measure of credit risk. The Reserve Bank has identified the external credit rating agencies that meet the eligibility criteria specified under the revised Framework.

Banks may rely upon the **ratings assigned by the external credit rating agencies chosen by the Reserve Bank** for assigning risk weights for capital adequacy purposes as per the mapping furnished in the RBI Guidelines.

Capital Charge for Credit Risk

The Circular issued by the Reserve Bank of India has laid down detailed guidelines on the capital adequacy requirements and the risk weights to be applied in case of the following claims:

- Claims on Domestic Sovereigns
- Claims on Foreign Sovereigns
- Claims on Public Sector Entities (PSES)
- Claims on Multilateral Development Banks, Bank for International Settlements and the International Monetary Fund
- Claims on Banks (Exposure to capital instruments)
- Claims on Primary Dealers
- Claims on Corporates, Asset Finance Companies and Non-Banking Finance
- Companies- Infrastructure Finance Companies
- Claims included in the Regulatory Retail Portfolios

Capital Charge for Credit Risk

- **Claims secured by Residential Property**
- **Claims Classified as Commercial Real Estate Exposure**
- **Non-Performing Assets (NPAs)**
- **Specified Categories Venture Capital Funds**
- **Other Assets like loans and advances to bank's own staff**
- **Off-Balance Sheet Items**
- **Securitisation Exposures**
- **Capital Adequacy Requirement for Credit Default Swap (CDS) Positions in the Banking Book**

Eligible Credit Rating Agencies

- ✓ Reserve Bank has undertaken the detailed **process of identifying the eligible credit rating agencies**, whose ratings may be used by banks for assigning risk weights for credit risk.
- ✓ In line with the provisions of the Revised Framework, where the facility provided by the bank possesses rating assigned by an eligible credit rating agency, the risk weight of the claim will be based on this rating.
- ✓ In accordance with the principles laid down in the Revised Framework, the Reserve Bank of India has decided that banks may use the ratings of the following domestic credit rating agencies for the purposes of risk weighting their claims falling under Corporate exposures for capital adequacy purposes.

Credit Rating Agencies

- Brickwork Ratings India Pvt. Limited (Brickwork);
- Credit Analysis and Research Limited;
- CRISIL Limited;
- ICRA Limited;
- India Ratings and Research Private Limited (India Ratings)
- SMERA Ratings Ltd. (SMERA)
- INFORMERICS Valuation and Rating Pvt. Ltd (from June, 2017)

The Reserve Bank of India has decided that banks may use the ratings of the following international credit rating agencies (arranged in alphabetical order) for the purposes of risk weighting their claims for capital adequacy purposes where specified:

- Fitch;
- Moody's; and
- Standard & Poor's

Internal Rating Based Approach

One of the most innovative aspects of the New Accord is the IRB approach to measurement of capital requirements for credit risk.

The IRB Approach offers the following two options: Foundation IRB Approach (FIRB) and Advanced IRB Approach (AIRB) version.

Since the approach is based on banks' internal assessments, the potential for more risk-sensitive capital requirements is substantial.

Internal Rating Based Approach

- **Probability of Default (PD),** which measures the likelihood that the borrower will default over a time given horizon.
- **Loss Given Default (LGD),** which measures the proportion of the exposure that will be lost if a default occurs.
- **Exposure At Default (EAD),** which for loan commitment measures the amount of the facility that is likely to be drawn in the event of a default.
- **Maturity (M),** which measures the remaining economic maturity of the exposure.

UNIT 14 Market Risk

STRUCTURE

Market Risk

Market risk is the risk of adverse deviations of the mark-to-market value of the trading portfolio, due to market movements, during the period required to liquidate the transactions.

The period of liquidation is critical to assess such adverse deviations. If the period of liquidation of the position gets longer, the possibilities of larger adverse deviations from the current market value also increase.

Trading Book

Banks also have several activities and undertake transactions that result in market exposure. They are not immune to these risks and have to face them too. All such transactions are reflected in the trading book.

A trading book consists of a bank's proprietary positions in financial instruments covering-

- **Debt Securities**
- **Equity**
- **Foreign Exchange**
- **Commodities (not permitted in our country presently)**
- **Derivatives held for Trading**

Trading Book

- ✓ The trading book also includes positions in financial instruments arising from matched principal brokering and market making, or positions taken in order to hedge other elements of the trading book.
- ✓ The proprietary positions are held with trading intent and with the intention of benefiting in the short-term, from actual and/or expected differences between their buying and selling prices or hedging other elements in the trading book.
- ✓ A bank's trading book exposure has the following risks, which arise due to adverse changes in the market variables such as interest rates, currency exchange rate, Commodity prices, market liquidity, etc., and their volatilities impact the bank's earnings and capital adversely.

Market Risk

- Liquidity Risk
  - Asset Liquidity Risk
  - Market Liquidity Risk
- Credit and Counterparty risks

Trading Liquidity Risk

Trading liquidity is the ability to freely transact in markets at reasonable prices. Trading liquidity is ability to liquidate positions without –

- **Affecting market prices**
- **Attracting the attention of other market participants.**

Trading Liquidity Risk

Liquidation risk arise from lack of trading liquidity and results in,

- **Adverse change in market prices**
- **Inability to liquidate position at a fair market price**
- **Large price changes caused by liquidation of position**
- **Inability to liquidate position at any price**

Credit & Counterparty Risk

Markets value the credit risk of issuers and borrowers and it reflects in prices. Credit risk of traded debts, such as bonds and debentures and commercial papers, etc., is indicated by Credit Rating given by rating agencies.

Credit rating indicates the risk level associated with the instruments and is factored into as add-ons to the risk-free rate of the corresponding maturity.

The lower the risk level, the lower is the spread over risk-free rate.

Market Risk Management Framework

Management processes for market risk management are designed essentially to answer these questions. Accordingly, management processes are sub-divided into the following four parts:

- Risk Identification
- Risk Measurement
- Risk Monitoring and Control
- Risk Mitigation

Organizational Structure

Usually, Market Risk Management organisation would consist:

- **The Board of Directors**
- **The Risk Management Committee**
- **The Asset-Liability Management Committee (ALCO)**
- **The ALM Support Group/Market Risk Group**
- **The Middle Office**

Organizational Structure

The Risk Management Committee is a Board level Sub-Committee. The responsibilities of Risk Management Committee with regard to market risk management aspects include the following:

- Setting guidelines for market risk management and reporting
- Ensuring that market risk management processes conform to the policy
- Setting up prudential limits and their periodical review
- Ensuring robustness of measurement of risk models

Organizational Structure

The Asset-Liability Management Committee (ALCO) is responsible for implementation of risk and business policies simultaneously in a consistent manner and decides on the business strategy to achieve these objectives. **Its role encompasses the following:**

- Product pricing for deposits and advances
- Maturity profile and mix of incremental assets and liabilities
- Articulating interest rate view of the bank
- Funding policy
- Transfer pricing
- Balance sheet management It set

Risk Identification

- ✓ All products and transactions should be analysed for risks associated with them. While, various risks associated with a standardised product stand analyzed, the risks in case of a non-standard products need to be analysed.
- ✓ Usually all standard products would have 'Product Programme' for each of them. All Risk- Taking Units operate within an approved Product Programme'.
- ✓ Product programme defines procedures, limits and controls for all aspects of the product. The product programme also specifies market risk measurement at an individual product level and at aggregate portfolio level.
- ✓ New products or non-standard products may operate under a 'Product Transaction Memorandum' on a temporary basis while a full Market Risk Product programme is being prepared.

Risk Measurement

Market risk measures are based on;

- **Sensitivity**
- **Downside Potential**

Sensitivity

Sensitivity, as had been stated deviation of market price due to unit movement of a single market parameter.

Supply-demand position, interest rate, market liquidity, inflation, exchange rate, stock prices, etc., are the market parameters, which drive market values.

Downside Potential

- **Risk materializes only when earnings deviate adversely. Downside potential captures the possible losses only and ignores the profit potential.** Downside risk is the most comprehensive measure of risk as integrates sensitivity and volatility with the adverse effect of uncertainty.
- This is the measure that is **most relied upon by banking and financial service industry as also the regulator.**

Risk Monitoring and Control

- **Policy guidelines limiting roles and authority**
- **Limit structure and approval process**
- **System and procedures to unbundle products and transactions to capture all risks**
- **Guidelines on portfolio size and mix**
- **Defined policy for mark-to-market**
- **Performance Measurement and Resource Allocation**

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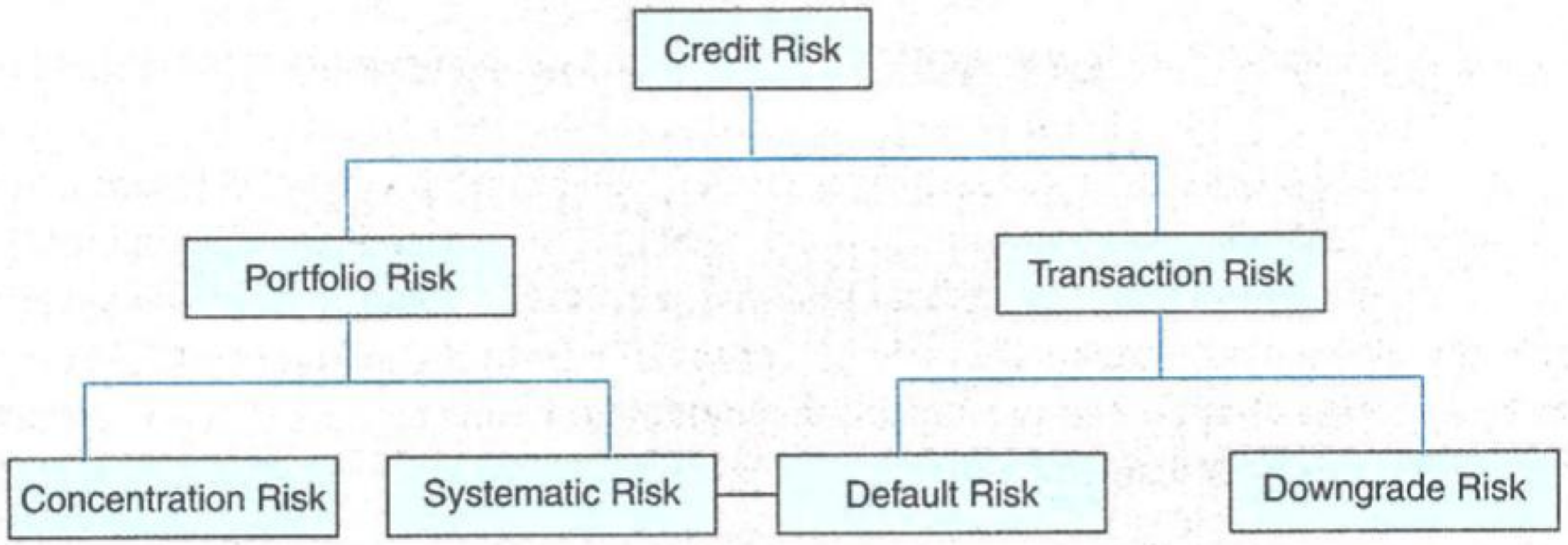
UNIT 15 Credit Risk

STRUCTURE

Credit Risk

Credit Risk is most simply defined as the potential of a bank borrower or counterparty fail to meet its obligations in accordance with agreed terms.

For most banks, loans and corporate bonds are the largest and most obvious source of credit risk.


Credit Risk
Portfolio Risk
Transaction Risk
Concentration Risk
Systematic Risk
Default Risk
Downgrade Risk

Credit Risk Management

Management processes are designed essentially to answer these questions. Accordingly, credit risk management processes are sub-divided into following four parts.

- Credit Risk Identification
- Credit Risk Measurement
- Credit Risk Monitoring and Control
- Credit Risk Mitigation Management of credit risk needs an organisation structure
- in place that can carry out the functions required for the purpose.

Organization Structure

Usually, Credit Risk Management organisation would consist of:

- **The Board of Directors**
- **The Risk Management Committee**
- **Credit Policy Committee (CPC)**
- **Credit Risk Management Department**

The Board of Directors

The Board of Directors has the overall responsibility for management of risks. The Board articulates credit risk management **policies, procedures, aggregate risk limits, review mechanisms and reporting and auditing systems.**

The Risk Management Committee

The Risk Management Committee is a Board level Sub-Committee. The responsibilities of Management Committee with regard to credit risk management aspects include the following:

- Setting guidelines for credit risk management and reporting
- Ensuring that credit risk management processes conform to the policy
- Setting up prudential limits and their periodical review
- Ensuring measurement of risk models
- Ensuring proper manning for the processes Management

Credit Policy Committee

Credit Policy Committee (CPC), also called Credit Control Committee/Credit Risk Management Committee (CRMC) deals with issues relating to credit policy and procedures and to analyse, manage and control credit risk on a bank wide basis. The Committee formulates policies on standards for presentation of credit proposals, financial covenants, rating standards and benchmarks, delegation of credit approving powers, prudential limits on large credit exposures

Credit Risk Management Department

Credit Risk Management Department (CRMD), which is independent of the Credit Administration Department, **enforces and monitors compliance of the risk parameters** and prudential limits set by the CPC/CRMC.

Department undertakes portfolio evaluations and conducts comprehensives studies on the environment to test the resilience of the loan portfolio.

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Risk Identification

Credit risk arises from potential changes in the credit quality of a borrower.

It has two components:

1. **Default risk**
2. **Credit spread risk**

Default Risk

Default risk is driven by the potential failure of a borrower to make promised payments, either partly or wholly.

In the event of default, a fraction of the obligations will normally be paid.

Credit Spread Risk

If a borrower does not default, there is still risk due to worsening in credit quality. This results in the possible widening of the credit-spread.

Usually this is reflected through rating downgrade. It is normally firm-specific.

Portfolio risk has two components

- Systematic or Intrinsic Risk
- Risk Concentration Risk

Systematic or Intrinsic Risk

Portfolio risk can be reduced through diversification. If a portfolio is fully diversified, i.e. diversified across geographies, industries, borrowers, markets, etc., equitably, then the portfolio risk is reduced to a minimum level. This minimum level corresponds to the risks in the economy which it is operating. This is systematic or intrinsic risk.

Concentration Risk

If the portfolio is not diversified that is to say that it has higher weight in respect of a borrower or geography or industry, etc., the portfolio gets concentration risk. A portfolio is open to the systematic risk i.e., the risks associated with the economy. If economy as a whole does not perform well, the portfolio performance will be affected.

Credit Rating

Credit Rating of an account is done with the primary objective to determine whether the account, after the expiry of a given period, would remain a performing asset, i.e., it will continue to meet its obligation to its creditors, including Bank and would not be in default.

A Credit Rating depicts **the credit quality of the borrower and depicts his default.** A credit rating process normally would consist of the following parameters:

- **Financial Parameter**
- **Management Parameter**
- **Industry Parameter**
- **Business Parameter**

Credit Risk Control & Monitoring

An elaborate and well-communicated policy at transaction level that articulates guidelines for risk taking, procedural **guidelines and an effective monitoring** system is necessary. This is also necessary to achieve the desired portfolio. Active portfolio management is required to keep up with the dynamics of the economy.

Consequently, **credit risk control and monitoring** is directed both at transaction level and portfolio level.

It must be mentioned here that an appropriate credit information system is the basic prerequisite for effective control and monitoring. A comprehensive and detailed **MIS (Management Information System) and CIS (Credit Information System)** is the backbone for an effective CRM System.

Policies & Guidelines at Transaction Level

The instruments of Credit Risk Management at transaction level are:

- **Credit Appraisal Process**
- **Risk Analysis Process**
- **Credit Audit and Loan Review**
- **Monitoring Process**

Policies & Guidelines at Transaction Level

Credit risk taking policy and guidelines at transaction level should be clearly articulated in the Bank's Loan Policy Document approved by the Board. Standards and guidelines should be outlined for

- **Delegation of Powers**
- **Powers Credit Appraisals**
- **Rating Standards and Benchmarks (derived from the Risk Rating System)**
- **Pricing Strategy**
- **Loan Review Mechanism**

Credit Approving Authority

Each Bank should have a carefully formulated scheme of delegation of powers. The banks should also evolve multi-tier credit approving system where the loan proposals are approved by **an 'Approval Grid'** or a Committee'. The 'Grid' or 'Committee', comprising at least 3 or 4 officers, may approve the credit facilities above a specified limit and invariably one officer should represent the CRMD, who has no volume and profit targets.

Controlling Credit Risk Through Loan Review Mechanism (LRM)

LRM is also called as Credit Audit. LRM an effective tool for constantly evaluating the quality of loan book and to bring about qualitative improvements in credit administration. Loan Review Mechanism is used for large value accounts with responsibilities assigned in various areas such as, **evaluating effectiveness of loan administration, maintaining the integrity of credit grading process, assessing portfolio quality, etc.**

Objectives of Loan Review Mechanism (LRM)

- To promptly identify loans, which develop credit weaknesses and initiate timely corrective action.
- To evaluate portfolio quality and isolate potential problem areas.
- To provide information for determining adequacy of loan loss provision.
- To assess the adequacy of and adherence to, loan policies and procedures, and to monitor compliance with relevant laws and regulations.
- To provide top management with information on credit administration, including credit sanction process, risk evaluation and post-sanction follow up.

Credit Derivativities

For most banks, particularly Indian banks, the single largest source of earnings and perhaps earnings volatility also are on account of credit risk. The traditional means to deal with credit risk include lending policies, credit approval processes, discretionary power structure, collateral and guarantees, concentration limits (with regard to single or group borrowers, industries or geographic regions), documentation, etc.

A credit derivative is an over-the-counter bilateral contract between two or more counterparties that provide for transfer of risks in a credit asset or credit portfolio without necessarily transferring the underlying asset from the books of the originator.

The reasons for entering into credit derivatives are:

Motives for Protection Buyers:

- Transferring credit risk without transferring the credit asset.
- Hedging against credit risks.
- Relief in regulatory capital required for credit assets which can be used for further business purpose.
- Better portfolio management by reducing concentration in market, i.e. diversification.

Motives for Protection Sellers:

- Yield Enhancement.
- Speculation.
- Arbitrage in case the credit derivative instruments are inefficiently priced.
- Diversification of credit risk.
- Credit derivatives initially were used only for hedging. However, during the last few years, credit derivatives are increasingly being used for speculative purposes also like any other derivative, viz., interest rate, equity and forex derivatives.

Users shall be classified by market-makers either as retail or non-retail for the purpose of offering credit derivative contracts. The following users shall be eligible to be classified as non-retail users:

a. NBFCs, including SPDs and HFCs, other than market-makers;
b. Insurance Companies regulated by Insurance Regulatory and Development Authority of India (IRDAI);
c. Pension Funds regulated by Pension Fund Regulatory and Development Authority (PFRDA);
d. Mutual Funds regulated by Securities and Exchange Board of India (SEBI);
e. Alternative Investment Funds regulated by Securities and Exchange Board of India (SEBI);
f. Resident companies with minimum net worth of Rs. 500 crore as per the latest audited balance sheet; and
g. Foreign Portfolio Investors (FPIs) registered with SEBI.

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UNIT 16 Operational Risk and Integrated Risk Management

STRUCTURE

Introduction

Operational risk is one area of risk that is faced by all organisations. The more complex an organisation is, the more would be its exposure to operational risk.

Operational risk would arise due to deviations from normal and planned functioning of systems, procedures, technology and human failures of omission and commission.

Operational risk is described by the Basel Committee as **"the risk of loss resulting from inadequate or failed internal processes, people, and systems, or from external events."** Accordingly, operational risk results from the possibility of suffering losses as a result of flaws or failures in an organization's internal procedures, employees' decisions, or technological infrastructure.

Nature of Operational Risk

- ✓ The Second Consultative Paper of Basel II suggested classification of operational risks based on the 'Causes' and 'Effects'.
- ✓ That is, classifications based on causes that are responsible for operational risks or classifications based on effects of risks were suggested. Classifications based on 'Causes' and 'Effects' are listed ahead.

Cause-Based

- **People oriented causes** - negligence, incompetence, insufficient training, integrity, key man.
- **Process oriented (Transaction based) causes** - business volume fluctuation, organizational complexity, product complexity, and major changes.
- **Process oriented (Operational control based) causes** - inadequate segregation of duties, lack of management supervision, inadequate procedures.
- ***Technology oriented causes*** - poor technology and telecom, obsolete applications, lack of automation, information system complexity, poor design, development and testing.
- ***External causes*** - natural disasters, operational failures of a third party, deteriorated social or political context.

Effect-Based

- Legal liability
- Regulatory, compliance and taxation penalties
- Loss or damage to assets
- Loss of recourse
- Write-downs

Event-Based

- Internal Fraud
- External Fraud
- Clients, products and business practices
- Damage to physical assets
- Business disruption and system failures
- Execution, delivery and process management

16.3 OPERATIONAL RISK CLASSIFICATION BY EVENT TYPE – DEFINITIONS

Internal Fraud: Losses due to acts of a type intended to defraud, misappropriate property or circumvent regulations, the law or company policy, excluding diversity/discrimination events, which involve at least one internal party.

External Fraud: Losses due to acts of a type intended to defraud, misappropriate property or circumvent the law, by a third party.

Employment Practices and Work Place Safety: Losses arising from acts inconsistent with employment, health or safety laws or agreements from payment of personal injury claims, or from diversity/discrimination events.

Clients, Products and Business Practices: Losses arising from an unintentional or negligent failure to meet a professional obligation to specific clients (including fiduciary and suitability requirements), or from the nature or design of a product.

Damage to Physical Assets: Losses arising from loss or damage to physical assets from natural disasters or other events.

Business Disruption and System Failures: Losses arising from disruption of business or system failures

Execution, Delivery and Process Management: Losses from failed transaction processing or process management, from relations with trade counterparties and vendors.

Operational Risk Management

17.6 OPERATIONAL RISK ORGANIZATIONAL FRAMEWORK

Ideally, the organizational set-up for operational risk management should include the following:

i. Board of Directors
ii. Risk Management Committee of the Board
iii. Operational Risk Management Committee
iv. Operational Risk Management Department
v. Operational Risk Managers
vi. Support Group for operational risk management

Fundamental principles of operational risk management

Principle 1

The board of directors should take the lead in establishing a strong risk management culture. The board of directors and senior management should establish a corporate culture that is guided by strong risk management and that supports and provides appropriate standards and incentives for professional and responsible behaviour.

Principle 2

Banks should develop, implement and maintain a Framework that is fully integrated into the bank's overall risk management processes. The Framework for operational risk management chosen by an individual bank will depend on a range of factors, including its nature, size, complexity and risk profile.

Principle 3

The board of directors should establish, approve and **periodically review the Framework.** The board of directors should oversee senior management to ensure that the policies, processes and systems are implemented effectively at all decision levels.

Principle 4

The board of directors should approve and review a risk appetite and tolerance statement for operational risk that articulates the nature, types, and levels of operational risk that the bank is willing to assume.

Governance

Principle 5

Senior management should develop for approval by the board of directors a clear, effective and robust governance structure with well defined, transparent and consistent lines of responsibility.

Principle 6

Senior management should ensure the identification and assessment of the operational **risk inherent in all material products/activities, processes and systems** to make sure the inherent risks and incentives are well understood.

Risk Management Environment

Principle 7

Senior management should ensure that there is an approval process for all new products, activities, processes and systems that fully assesses operational risk.

Principles 8

Senior management should implement a process to regularly monitor operational risk profiles and material exposures to losses.

Appropriate reporting mechanisms should be in place at the board, senior management, and business line levels that support proactive management of operational risk.

Control and Mitigation

Principle 9

Banks should have a **strong control environment that utilises policies, processes and systems**; appropriate internal controls; and appropriate risk mitigation and/or transfer strategies.

Principle 10

Banks should have business resiliency and continuity plans in place to ensure an ability to operate on **an ongoing basis and limit losses** in the event of severe business disruption.

Principle 11

Operational Risk Management Practices should be based on a **well laid out policy duly approved at the board level** that describes the processes involved in controlling operational risks

Business Resiliency and Continuity

The policy should cover

- **Operational risk management structure**
- **Role and responsibilities**
- **Operational risk management processes**
- **Operational risk assessment/measurement methodologies**

1. **Internal Fraud**

Losses due to acts of a type intended to defraud, misappropriate property, or circumvent regulations, the law, or company policy, excluding diversity/ discrimination events, which involves at least one internal party

Categories (Level 2)	Activity Examples (Level 3)
Unauthorized Activity	Transactions not reported (intentional) Transaction type unauthorized (w/monetary loss) Mismarking of position (intentional)
Theft and Fraud	Fraud/credit fraud/worthless deposits Theft/extortion/embezzlement/robbery Misappropriation of assets Malicious destruction of assets Forgery Check kiting Smuggling Account takeover/impersonation/etc. Tax noncompliance/evasion (willful) Bribes/kickbacks Insider trading (not on firm's account)

2. External Fraud

Losses due to acts of a type intended to defraud, misappropriate property, or circumvent the law, by a third party.

Categories (Level 2)	Activity Examples (Level 3)
Theft and Fraud	Theft/Robbery Forgery Check kiting
Systems Security	Hacking damage Theft of information (w/monetary loss)

3. Employment Practices and Workplace Safety

Losses arising from acts inconsistent with employment, health, or safety laws or agreements, from payment of personal injury claims, or from diversity/discrimination events.

Categories (Level 2)	Activity Examples (Level 3)
Employee Relations	Compensation, benefit, termination issues Organized labor activity
Safe Environment	General liability (slip and fall, etc.) Employee health and safety rules events Workers' compensation
Diversity and Discrimination	All discrimination types

4. **Clients, Products, and Business Practices**

Losses arise from an unintentional or negligent failure to meet a professional obligation to specific clients (including fiduciary and suitability requirements), or from the nature or design of a product.

Suitability, Disclosure, and Fiduciary	Fiduciary breaches/guideline violations Suitability/disclosure issues (KYC, etc.) Retail customer disclosure violations Breach of privacy Aggressive sales Account churning Misuse of confidential information Lender liability
Improper Business or Market Practices	Antitrust Improper trade/market practices Market manipulation Insider trading (on firm's account) Unlicensed activity Money laundering
Product Flaws	Product defects (unauthorized, etc.) Model errors
Selection, Sponsorship, and Exposure	Failure to investigate client per guidelines Exceeding client exposure limits
Advisory Activities	Disputes over performance of advisory activities

5. Damage to Physical Assets

Losses arising from loss or damage to physical assets from natural disaster or other events.

Categories (Level 2)	Activity Examples (Level 3)
Disasters and Other Events	Natural disaster losses Human losses from external sources (terrorism, vandalism)

6. Business Disruption and System Failures

Losses arising from disruption of business or system failures.

Categories (Level 2)	Activity Examples (Level 3)
Systems	Hardware Software Telecommunications Utility outage/disruptions

16.14 INTEGRATED RISK MANAGEMENT – APPROACH

The process of Integrated Risk Management consists of

- *Strategy:* Integration of risk management as a key corporate strategy.
- *Organisation:* Establishment of the Chief Risk Officer position with his/her accountability to the board of directors. As part of effective risk management, RBI has advised banks, inter-alia, to have a system of separation of credit risk management function from the credit sanction process. However, RBI has observed that the banks follow diverse practices in this regard. In order to bring uniformity in approach followed by banks, as also, to align the risk management system with the best practices, RBI has advised Banks vide its guidelines of April, 2017 as under:
  - Banks should appoint a Chief Risk Officer.
  - Banks should lay down a Board-approved policy clearly defining the role and responsibilities of the CRO.
  - CRO has to control and co-ordinate the functions of the Risk Management Department of the Bank.
  - Because of the importance RBI gives to this post, RBI has made CRO to report directly to the MD or CEO of the Bank.
- *Process:* The process of identifying, assessing, controlling and financing risk must be common across the whole enterprise.
- *Systems:* Risk management systems must be developed to provide information and analytical tools to support the Enterprise Risk Management functions.

When properly implemented, *Integrated Risk Management:*

- Aligns the strategic aspects of risk with day-to-day operational activities.
- Facilitates greater transparency for investors and regulators.
- Enhances revenue and earnings growth.
- Controls downside risk potential.

Implementation of Integrated Risk Management has challenges. They are its Real-Time concern, business Challenges and cultural issues. The Process of Integrated Risk Management consists of

- ***Strategy:*** Integration of risk management as a key corporate strategy.
- ***Organisation:*** Establishment of the Chief Risk Officer's position with his/her accountability to the board of directors.
- *Process:* The process of identifying, assessing, controlling and financing risk must be common across the whole enterprise.
- *Systems:* Risk management systems must be developed to provide information and analytical tools to support the Enterprise Risk Management functions.

RBI has advised Banks vide its guidelines of April, 2017 as under:

- Banks should appoint a Chief Risk Officer.
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- CRO has to control and co-ordinate the functions of the Risk Management Department of the Bank.
- Because of the importance RBI gives to this post, RBI has made CRO to report directly to the MD or CEO of the Bank.

UNIT 17

Liquidity Risk Management

STRUCTURE

Introduction

A bank is said to be solvent if it's net worth is not negative. To put it differently, a bank is solvent if the total realizable value of its assets is more than its outside liabilities (i.e., other than its equity/owned funds).

Liquidity Risk

As such, at any point in time, a bank could be;

(i) both solvent and liquid or

(ii) liquid but not solvent or

(iii) solvent but not liquid or

(iv) neither solvent nor liquid.

Potential Liquidity Risk Drivers

17.3 POTENTIAL LIQUIDITY RISK DRIVERS

The internal and external factors in banks that may potentially lead to liquidity risk problems in Banks are as under:

Internal Banking Factors	*External Banking Factors*
High off-balance sheet exposures.	Very sensitive financial market depositors.
The banks rely heavily on the short-term corporate Deposits/wholesale deposits.	External and internal economic shocks.
A negative gap (liability is more than the asset) in the maturity dates of assets and liabilities.	Low/slow economic performances.
The banks' rapid asset expansions exceed the available funds on the liability side.	Decreasing depositors' trust on the banking sector.
Concentration of deposits in the short-term Tenor.	Non-economic factors
Less allocation in the liquid government instruments.	Sudden and massive liquidity withdrawals from depositors.
Fewer placements of funds in long-term deposits.	Unplanned termination of government deposits.

Key Consideration in Liquidity Risk

- Availability of liquid assets,
- Extent of volatility of the deposits,
- Degree of reliance on volatile sources of funding,
- Level of diversification of funding sources,
- Historical trend of stability of deposits,
- Quality of maturing assets,
- Market reputation,
- Impact of off balance sheet exposures on the balance sheet, and
- Contingency plans.

Types of Liquidity Risk

- **Funding Liquidity Risk**
- **Market liquidity Risk**

Funding Liquidity Risk - the risk that a bank will not be able to meet efficiently the expected and unexpected current and future cash flows and collateral needs without affecting either its daily operations or its financial condition.

Market liquidity Risk - the risk that a bank cannot easily offset or eliminate a position at the prevailing market price because of inadequate market depth or market disruption.

Liquidity Risk Management Principles

After the global financial crisis, in recognition of the need for banks to improve their liquidity risk management, the Basel Committee on Banking Supervision **(BCBS) published "Principles for Sound Liquidity Risk Management and Supervision**" in September 2008.

The broad principles for sound liquidity risk management by banks as envisaged by BCBS are as under following principles;

Principle 1

A bank is responsible for **the sound management of liquidity risk**. A bank should establish a robust liquidity risk management framework that ensures it maintains **sufficient liquidity, including high quality liquid assets, to withstand a range of stress events**, including those involving the loss or impairment of both unsecured and secured funding sources.

Principle 2

A bank should clearly **articulate a liquidity risk tolerance that is appropriate for its business strategy** and its role in the financial system.

Principle 3

Senior management should **develop a strategy, policies and practices** to manage liquidity risk in accordance with the risk tolerance and to ensure that the bank maintains sufficient liquidity.
Senior management should continuously review information on the bank's liquidity developments and report to the board of directors on a regular basis.
A bank's board of directors should **review and approve the strategy, policies and practices related to the management of liquidity** at least annually and ensure that senior management manages liquidity risk effectively.

Principle 4

A bank should incorporate liquidity costs, benefits and risks in the **internal pricing, performance measurement and new product approval process for all significant business activities** (both on- and off-balance sheet), thereby aligning the risk-taking incentives of individual business lines with the liquidity risk exposures their activities create for the bank as a whole.

Principle 5

A bank should have a **sound process for identifying, measuring, monitoring and controlling liquidity risk.** This process should include a robust framework for comprehensively projecting cash flows arising from assets, liabilities and off-balance sheet items over an appropriate set of time horizons.

Principle 6

A bank should actively monitor and control liquidity **risk exposures and funding needs within and across legal entities**, business lines and currencies, taking into account legal, regulatory and operational limitations to the transferability of liquidity.

Principle 7

A bank should **establish a funding strategy** that provides effective diversification in the sources and tenor of funding. It should maintain an ongoing presence in its chosen funding markets and strong relationships with funds providers to promote effective diversification of funding sources.

Principle 8
A bank should actively manage its intraday liquidity positions and risks to meet payment and settlement obligations on a timely basis under both normal and stressed conditions and thus contribute to the smooth functioning of payment and settlement systems.

Principle 9

A bank should actively manage its **collateral positions, differentiating between encumbered and unencumbered assets**. A bank should monitor the legal entity and physical location where collateral is held and how it may be mobilised in a timely manner.

Principle 10

A bank should **conduct stress tests on a regular basis** for a variety of short-term and protracted institution-specific and market-wide stress scenarios (individually and in combination) to identify sources of potential liquidity strain and to ensure that current exposures remain in accordance with a bank's established liquidity risk tolerance.

Principle 11

A bank should have a **Formal Contingency Funding Plan (CFP)** that clearly sets out the strategies for addressing liquidity shortfalls in emergency situations.
A CFP should outline policies to manage a range of stress environments, establish clear lines of responsibility, include clear invocation and escalation procedures and be regularly tested and updated to ensure that it is operationally robust.

Principle 12

A bank should maintain a cushion of unencumbered, high quality liquid assets to be held as insurance against a range of liquidity stress scenarios, including those that involve the loss or impairment of unsecured and typically available secured funding sources. There should be no legal, regulatory or operational impediment to using these assets to obtain funding.

Principle 13

A bank should publicly disclose information on a regular basis that enables market participants to make an informed judgment about the soundness of its liquidity risk management framework and liquidity position.

Governance of Liquidity Risk Management

- Board of Directors
- Risk Management Committee
- Assets Liability Management Committee
- Assets Liability Management Support Group

Board of Directors

The BoD should have the overall responsibility for management of liquidity risk.

The Board should decide the strategy, policies and procedures of the bank to manage liquidity risk in accordance with the liquidity risk tolerance/limits

The Risk Management Committee

The Risk Management Committee, which reports to the Board, consisting of Chief Executive Officer (CEO) Chairman and Managing Director (CMD) and heads of credit, market and operational risk management committee should be responsible for **evaluating the overall risks faced by the bank including liquidity risk.**

The Asset-Liability Management Committee (ALCO)

The Asset-Liability Management Committee (ALCO) consisting of the bank's top management should be responsible for **ensuring adherence to the risk tolerance/limits set by the Board** as well as implementing the liquidity risk management strategy of the bank in line with bank's decided risk management objectives and risk tolerance.

The Asset Liability Management (ALM) Support Group:

The ALM Support Group consisting of operating staff should be responsible for analysing, monitoring and reporting the liquidity risk profile to the ALCO.

The group should also prepare **forecasts (simulations) showing the effect of various possible changes in market conditions** on the bank's liquidity position and recommend action needed to be taken to maintain the liquidity position/adhere to bank's internal limits.

Liquidity Risk Management Policy

- **Liquidity Risk Tolerance**
- **Strategy for Managing Liquidity Risk**

Liquidity Risk Tolerance

Banks should have an explicit liquidity risk tolerance set by the Board of Directors.

The risk tolerance should define the level of liquidity risk that the bank is willing to assume, and should reflect the bank's financial condition and funding capacity.

Strategy for Managing Liquidity Risk

The strategy for managing liquidity risk should be appropriate for the nature, scale and complexity of a bank's activities. In formulating the strategy, banks/banking groups should take into consideration its **legal structures, key business lines, the breadth and diversity of markets, products, jurisdictions** in which they operate and home and host country regulatory requirements, etc.

Topic: Liquidity Risk Management

- **Risk Identification**
- **Measurement of Liquidity Risk**

Risk Identification

A bank should **define and identify the liquidity risk** to which it is exposed for each major on and off balance sheet position, including the effect of embedded options and other contingent exposures that may affect the bank's sources and uses of funds and for all currencies in which a bank is active.

Measurement

There are two simple ways of measuring liquidity; **one is the stock approach and the other, flow approach**.

The stock approach, certain ratios, like liquid assets to short term total liabilities, purchased funds to total assets, core deposits to total assets, loan to deposit ratio, etc., are calculated and compared to the benchmarks that a bank has set for itself. While the stock approach helps up in looking at liquidity from one angle, it does not reveal the intrinsic liquidity profile of a bank.

Ratio Analysis

Sl. No.	Ratio	Significance	Industry Average (in %)
1.	**(Volatile liabilities – Temporary Assets)/ (Earning Assets – Temporary Assets)**	Measures the extent to which volatile money supports bank's basic earning assets. Since the numerator represents short-term, interest sensitive funds, a high and positive number implies some risk of illiquidity.	40
2.	**Core deposits/Total Assets**	Measures the extent to which assets are funded through stable deposit base.	50
3.	**(Loans + mandatory SLR + mandatory CRR + Fixed Assets)/ Total Assets**	Loans including mandatory cash reserves and statutory liquidity investments are least liquid and hence a high ratio signifies the degree of 'illiquidity' embedded in the balance sheet.	80
4.	**(Loans + mandatory SLR + mandatory CRR + Fixed Assets)/Core Deposits**	Measure the extent to which illiquid assets are financed out of core deposits.	150
5.	**Temporary Assets/Total Assets**	Measures the extent of available liquid assets. A higher ratio could impinge on the asset utilisation of banking system in terms of opportunity cost of holding liquidity.	40
6.	**Temporary Assets/Volatile Liabilities**	Measures the cover of liquid investments relative to volatile liabilities. A ratio of less than 1 indicates the possibility of a liquidity problem.	60
7.	**Volatile Liabilities/Total Assets**	Measures the extent to which volatile liabilities fund the balance sheet.	60

Ratio Analysis

Volatile Liabilities: (Deposits + borrowings and bills payable up to 1 year). Letters of credit – full outstanding. Component-wise CCF of other contingent credit and commitments. Swap funds (buy/sell) up to one year. Current deposits (CA) and Savings deposits (SA) i.e. (CASA) deposits reported by the hanks as payable within one year (as reported in structural liquidity statement) are included under volatile liabilities. Borrowings include from RBI, call, other institutions and refinance.

Temporary assets = Cash + Excess CRR balances with RBI + Balances with banks + Bills purchased/ discounted up to 1 year + Investments up to one year + Swap funds (sell/buy) up to one year.

Ratio Analysis

Earning Assets = Total assets - (Fixed assets + Balances in current accounts with other banks + Other assets excluding leasing + Intangible assets).

Core deposits = All deposits (including CASA) above 1 year (as reported in structural liquidity statement) + net worth.

Reporting to RBI

17.16 REPORTING TO THE RESERVE BANK OF INDIA

Banks are required to submit the liquidity return, as per the prescribed format to the Chief General Manager-in-Charge, Department of Banking Supervision, Reserve Bank of India, Central Office, World Trade Centre, Mumbai as detailed below:

Statement of Structural Liquidity:

The existing liquidity reporting requirements have been reviewed by RBI. Under the revised requirements, the time buckets are to be aligned as (with effect from February 1, 2016) as per the table given below:

Next day	2-7 days	8-14 days	15-30 days	31 day to 2 mon	>2 m to 3 mon	>3 m to 6 m	>6m to 1 yr	>1yr to 3 yrs	>3yr to 5 yrs	More than 5 yrs

Further, the statement is required to be reported in five parts viz.

(i) 'for domestic currency, Indian operations';
(ii) 'for foreign currency, Indian operations';
(iii) 'for combined Indian operations';
(iv) 'for overseas operations' and for
(v) 'consolidated bank operations'.

While statements at (i) to (iii) are required to be submitted fortnightly, statements at (iv) and (v) are required to be submitted at monthly and quarterly intervals, respectively. The periodicity in respect of each part of the return is given in the table ahead:

Sl. No.	Name of the Liquidity Return (LR)	Periodicity	Time period by which required to be reported
Structural Liquidity Statement			
(i)	Part A1 – Statement of Structural Liquidity – Domestic Currency, Indian Operations	Fortnightly*	within a week from thereporting date
(ii)	Part A2 – Statement of Structural Liquidity – Foreign Currency, Indian Operations	do	do
(iii)	Part A3 – Statement of Structural Liquidity – Combined Indian Operations	do	do
(iv)	Part B – Statement of Structural Liquidity for Overseas Operations	Monthly#	within 15 days from thereporting date
(v)	Part C – Statement of Structural Liquidity – ForConsolidated Bank Operations	Quarterly#	within a month from thereporting date

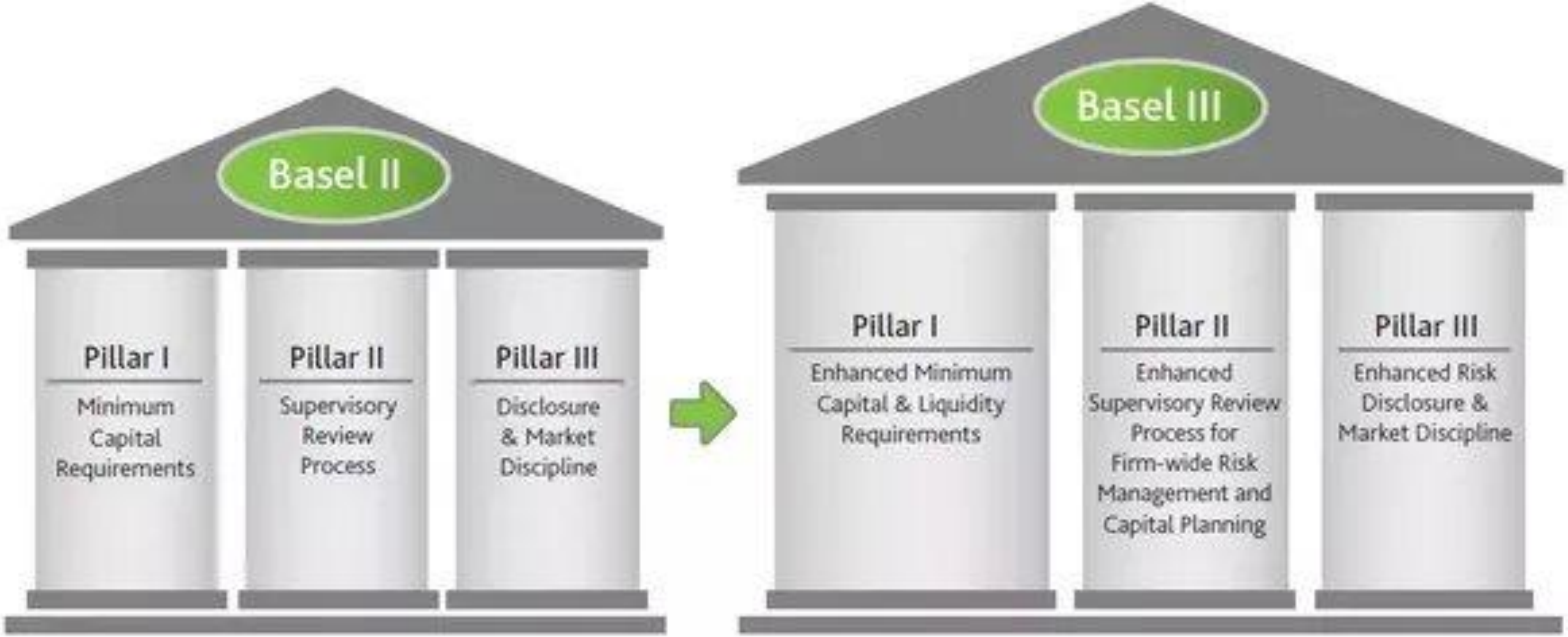
*Reporting dates will be 15th and last date of the month – in case these dates are holidays, the reporting dates will be the previous working day.

Reporting date will be the last working day of the month/quarter.

UNIT 18

Basel-III Framework on Liquidity Standards

STRUCTURE


Basel II
Pillar I
Minimum Capital Requirements
Pillar II
Supervisory Review Process
Pillar III
Disclosure & Market Discipline
Basel III
Pillar I
Enhanced Minimum Capital & Liquidity Requirements
Pillar II
Enhanced Supervisory Review Process for Firm-wide Risk Management and Capital Planning
Pillar III
Enhanced Risk Disclosure & Market Discipline

Basel III

Basel III or Basel 3 released in December, 2010 is the third in the series of Basel Accords. These accords deal with risk management aspects for the banking sector.

So we can say that **Basel III is the global regulatory standard on bank capital adequacy, stress testing and market liquidity risk.**

The RBI issued Guidelines based on the Basel III reforms on capital regulation on May 2 2012, to the extent applicable to banks operating in India.

The Basel III capital regulation has **been implemented form April 1, 2013** in India in phase and it will be fully implemented as on March 31, 2019 but Extended.

Capital Requirements under Basel III

Thus, with full implementation of capital ratios and CCB the capital requirements as on 1st October, 2021will be as follows:

S. No.	*Regulatory Capital*	*As % to RWAs*
(i)	Minimum Common Equity Tier 1 Ratio	5.5
(ii)	Capital Conservation Buffer (comprised of Common Equity)	2.5
(iii)	Minimum Common Equity Tier 1 Ratio plus Capital Conservation Buffer [(i)+(ii)]	8.0
(iv)	Additional Tier 1 Capital	1.5
(v)	Minimum Tier 1 Capital Ratio [(i) +(iv)]	7.0
(vi)	Tier 2 Capital	2.0
(vii)	Minimum Total Capital Ratio (MTC) [(v)+(vi)]	9.0
(viii)	Minimum Total Capital Ratio plus Capital Conservation Buffer [(vii)+(ii)]	11.5

18.2 LIQUIDITY COVERAGE RATIO (LCR)

The Liquidity Coverage ratio is computed as under:

$$\frac{\text{Stock of High-Quality Liquid Assets}}{\text{Total Net Cash Outflows over the next 30 calendar days}} \geq 100\%$$

The LCR requirement is binding on banks from January 1, 2015. However, to provide a transition time for banks, Reserve Bank of India has permitted a gradual increase in the ratio starting with a minimum 60% for the calendar year 2015 as per the time-line given below:

	January 1 2015	January 1 2016	January 1 2017	January 1 2018	January 1 2019
Minimum LCR	**60%**	**70%**	**80%**	**90%**	**100%**

Due to the pandemic situation, RBI relaxed the maintenance of LCR as below and reinstated the 100% requirement from April 1, 2021 onwards.

Period	***Per cent***
April 17, 2020 to September 30, 2020	80%
October 1, 2020 to March 31, 2021	90%
April 1, 2021 onwards	100%

The stress scenario specified by the BCBS entails a combined idiosyncratic (Bank-specific) and market-wide shock that would result in:

(a) the run-off of a proportion of retail deposits;
(b) a partial loss of unsecured wholesale funding capacity;
(c) a partial loss of secured, short-term financing with certain collateral and counter parties;
(d) additional contractual outflows that would arise from a downgrade in the bank's public credit rating by up to three notches, including collateral posting requirements;
(e) increases in market volatilities that impact the quality of collateral or potential future exposure of derivative positions and thus require larger collateral haircuts or additional collateral, or lead to other liquidity needs;
(f) unscheduled draws on committed but unused credit and liquidity facilities that the bank has provided to its clients; and
(g) the potential need for the bank to buy back debt or honour non-contractual obligations in the interest of mitigating reputational risk.

18.2.1 High Quality Liquid Assets

(a) Liquid assets comprise of high-quality assets that can be readily sold or used as collateral to obtain funds in a range of stress scenarios. They should be unencumbered i.e., without legal, regulatory or operational impediments. Assets are considered to be high quality liquid assets, if they can be easily and immediately converted into cash at little or no loss of value. The liquidity of an asset depends on the underlying stress scenario, the volume to be monetized and the timeframe considered. Nevertheless, there are certain assets that are more likely to generate funds without incurring large discounts due to fire-sales even in times of stress.

(b) While the fundamental characteristics of these assets include low credit and market risk; ease and certainty of valuation; low correlation with risky assets and listing on a developed and recognized exchange market, the market related characteristics include active and sizeable market; presence of

Banks, with effect January 1, 2015, are permitted to avail liquidity facility against such securities under a special facility called 'Facility to Avail Liquidity for Liquidity Coverage Ratio' (FALLCR), essential features of which are given below:

(i) **Eligibility:** Availing of liquidity against such securities would be permitted to banks only under the conditions of stress as described under paragraph 4.3 of the above-mentioned circular dated June 9, 2014, and after utilisation of all other HQLAs (including securities permitted under MSF). Banks will be required to furnish a declaration to this effect that they have exhausted their all other HQLAs before availing of the FALLCR.

(ii) **Tenor:** This facility can be availed/rolled over up to a maximum period of 90 days.

(iii) **Haircut:** Liquidity against securities under FALLCR will be available after applying haircuts as stipulated for MSF.

(iv) **Facility rate:** Rate of interest on the funds availed under this facility will be 200 bps above the prevailing LAF repo rate, up to a period of 90 days, or as decided by the RBI from time to time.

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Liquidity Risk Monitoring Tools

- Contractual Mismatch Maturity
- Concentration of Funding
- Available Unencumbered Assets
- LCR by Significant Currency
- Market-related Monitoring Tools
- Basel III Liquidity Returns

Basel III Liquidity Returns

18.3.1 Basel III Liquidity Returns

S. No.	Name of the Basel III Liquidity Return(BLR)	Frequency of Submission	Time period by which Required to be Reported
1.	Statement on Liquidity Coverage Ratio (LCR)-BLR-1	Monthly	within 15 days
2.	Statement of Funding Concentration – BLR-2	Monthly	within 15 days
3.	Statement of Available Unencumbered Assets – BLR-3	Quarterly	within a month
4.	LCR by Significant Currency – BLR-4	Monthly	within a month
5.	Statement on Other Information on Liquidity – BLR-5	Monthly	within 15 days

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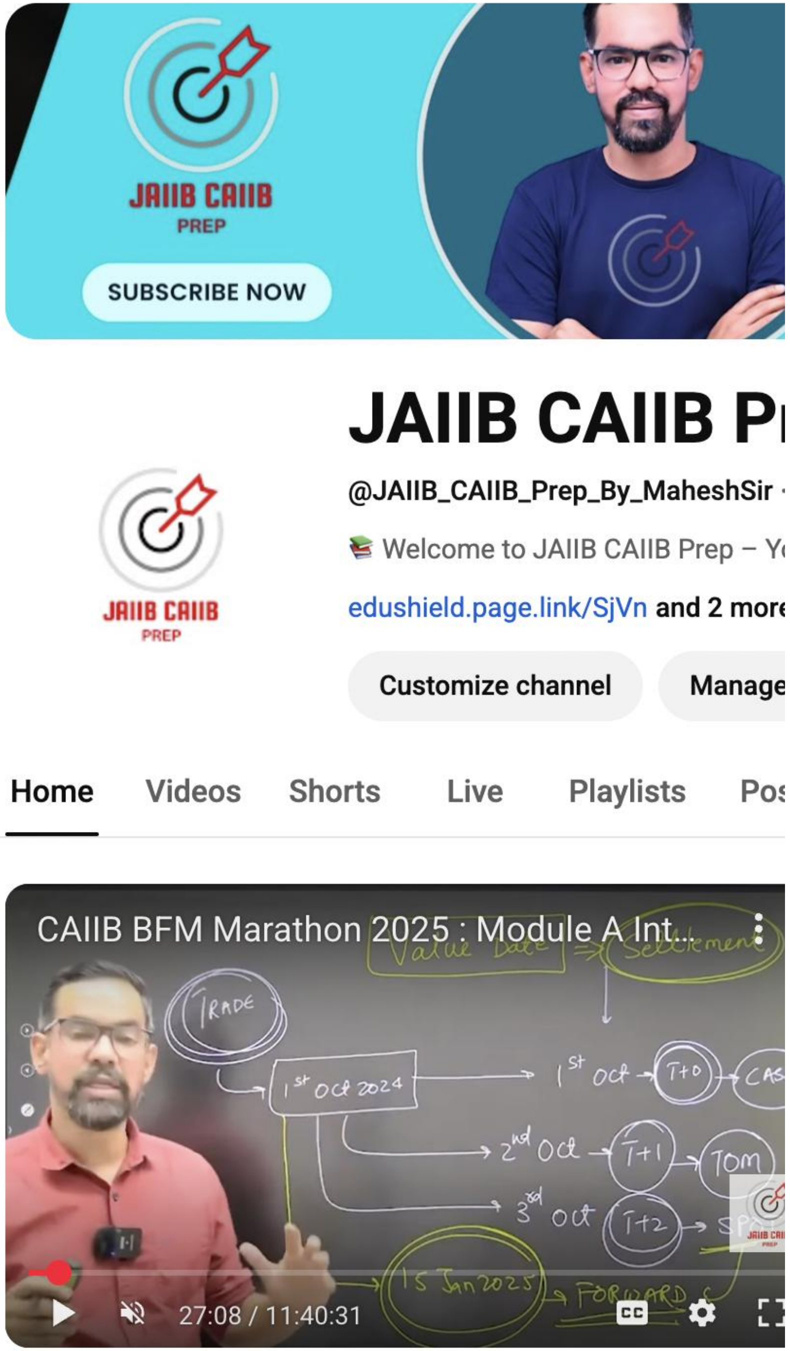

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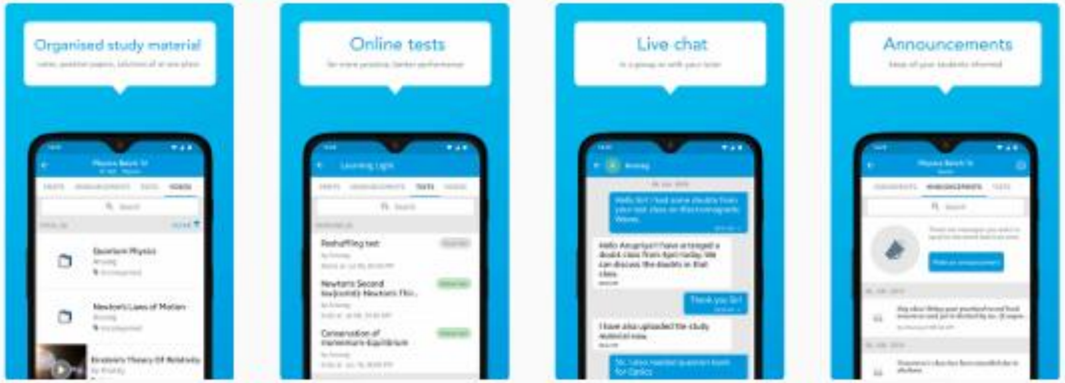

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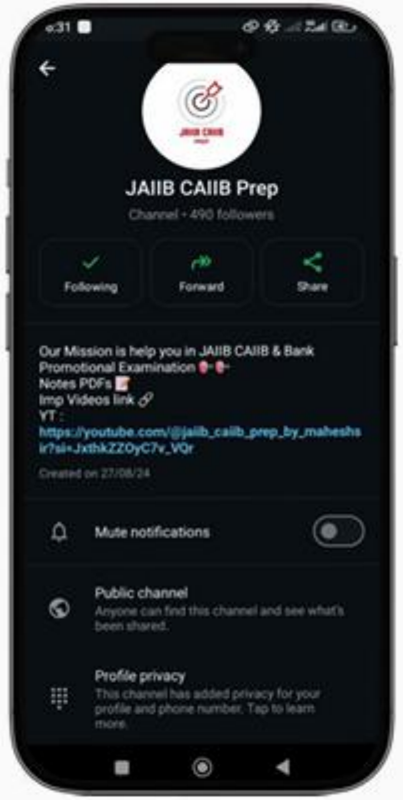

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